

(Revision-2021)

Reg.No

Signature.....

MODEL QUESTION PAPER (REV-21)

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

ARTIFICIAL INTELLIGENCE-SET-I

(Maximum Marks : 75)

TIME : 2HRS

PART -A

Answer all questions in one word or one sentence

I.

1. What is Artificial Intelligence?
2. List any two application of AI.
3. Differentiate between informed search and uninformed search.
4. Define Heuristic function.
5. List any two uninformed search strategies.
6. What are Logic agents?
7. Define knowledge based agents.
8. What are the components of a Constraint Satisfaction Problem?
9. Define First Order Predicate Logic .

(9x1=9
)

PART B

Answer any eight questions from the following. Each question carries 3 marks.

II.

1. Define an intelligent agent with an Example?
2. Discuss about history of AI.
3. Differentiate BFS and DFS search strategies.
4. Explain fully observable and partially observable environments.
5. Explain disadvantages of propositional logic .
6. Discuss any three real world problems?
7. Explain mini-max algorithm.
8. Define rational agent.
9. Differentiate propositional and first-order logic.
10. Explain 8-puzzle problem.

(8x3=24)

PART C

Answer all questions from the following. Each question carries 7 marks.

III

1. Describe any two ways to represent states and the transition between them in agent programs.

OR

2. Explain the structure Simple reflex agents and Model-based reflex agents
3. Explain the components of a well defined AI problem? Write the standard formulation of 8-queens problem.

OR

4. Describe any four uninformed search strategies
5. Solve the following crypt arithmetic problem.

$$\begin{array}{r} T \ W \ O \\ + \ T \ W \ O \\ \hline F \ O \ U \ R \end{array}$$

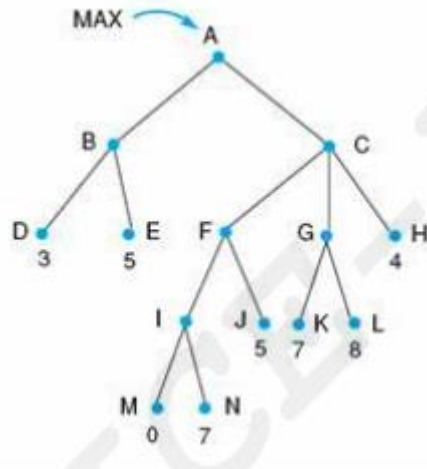
OR

6. Explain in detail about alpha-beta pruning
7. Convert the following sentences into first order logic:

Everyone who loves all animals is loved by someone.
Anyone who kills an animal is loved by no one.
Jack loves all animals.
Either Jack or Curiosity killed the cat, who is named Tuna.
Did Curiosity kill the cat?

OR

8. Explain in detail about Propositional logic.
9. Consider the following figure. Perform min-max procedure on the below given tree.



OR

10. Why knowledge representation is necessary in AI systems? Give AI systems in which knowledge is important?.
11. Solve the following crypt arithmetic problem stating all the constraints.

$$\begin{array}{r} \text{EAT} \\ + \text{THAT} \\ \hline \text{APPLE} \end{array}$$

OR

12. Differentiate between A* algorithm and Best First Search algorithm.
- 13.

OR

14. Differentiate between Depth limited search and DFS

(6x7=42)

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MODEL QUESTION PAPER (REV-21)

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

ARTIFICIAL INTELLIGENCE-SET-II

(Maximum Marks : 75)

TIME : 2HRS

PART -A

Answer all questions in one word or one sentence

I.

1. Discuss the history of AI?
2. Define Intelligent agent.
3. List any four real world problems
4. Define Heuristic function.
5. Define uniform cost search.
6. What is pruning?
7. Define knowledge based agents.
8. What are the components of a Constraint Satisfaction Problem?
9. Define propositional Logic .

(9x1=9
)

PART B

II. Answer any eight questions from the following. Each question carries 3 marks.

1. Define an intelligent agent with an Example?
2. List application of AI.
3. Differentiate uninformed and informed search strategies.
4. Explain Different types of environments.
5. Explain atomic and complex sentences in propositional logic .
6. Discuss any two toy-problems?
7. Explain heuristic search technique with example.
8. Define components of CSP problems.
9. Differentiate propositional and first-order logic.
10. What is the importance of two bounds in Alpha-Beta cut-offs.

(8x3=24)

PART C

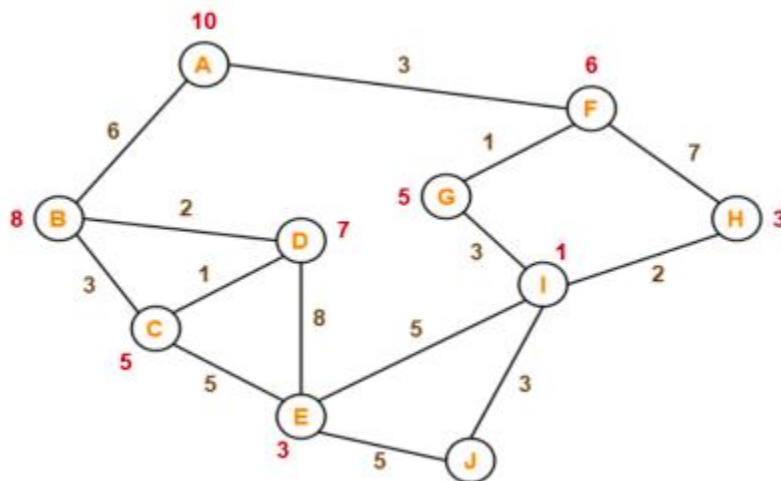
Answer all questions from the following. Each question carries 7 marks.

III

1. Define the component of 8 puzzle problem with suitable example.

OR

2. Explain the structure Goal based agents and Utility based agents
3. Find the most cost-effective path to reach from start state A to final state J using A*Algorithm.

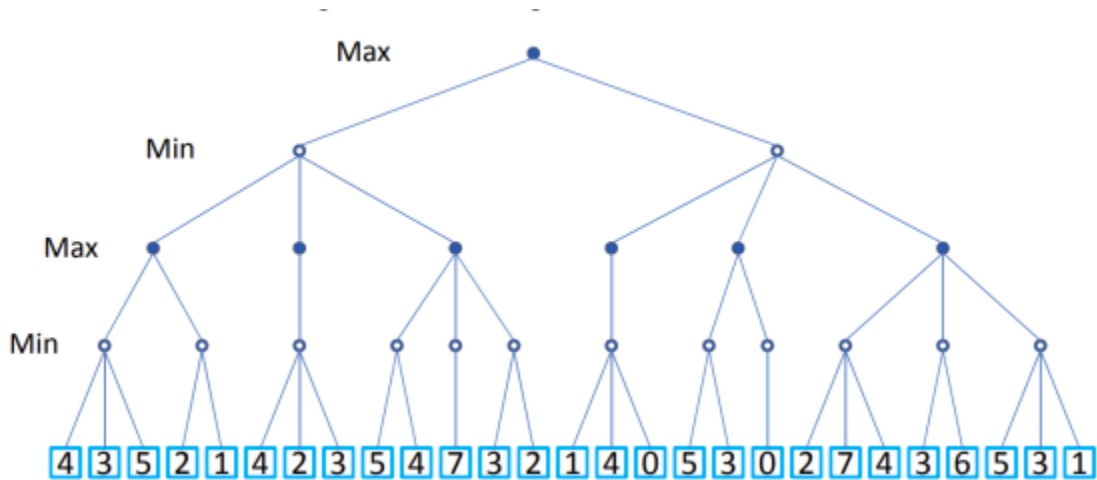


OR

4. Discuss in detail about Greedy best first search.
5. Solve the following crypt arithmetic problem.
$$\begin{array}{r} \text{LET} \\ + \text{LEE} \\ \hline \text{ALL} \end{array}$$

OR

6. Solve the below tree using Min-max algorithm.



7. Convert the following sentences into first order logic:

Everyone who loves all animals is loved by someone.
 Anyone who kills an animal is loved by no one.
 Jack loves all animals.
 Either Jack or Curiosity killed the cat, who is named Tuna.
 Did Curiosity kill the cat?

OR

8. Explain in detail about Knowledge based agents.

9. Explain in detail about alpha-beta pruning? What is the advantages of using pruning.

OR

10. Why knowledge representation is necessary in AI systems? Give AI systems in which knowledge is important?.

11. Explain min-max procedure in detail.

OR

12. Discuss real world problems of AI in detail.

13. Explain in detail about Task environment types.

OR

14. Explain heuristic search technique with example.

(6x7=42)